Aim: to investigate how particle size distribution in each pit affects the physical growth of plants population from each soil pit as we move down the catena

Objective: compare the change in %clay and %sand in horizon A of each soil pit
Look at how the Mean tree height and the maximum tree height, of trees around the three study sites changes, I will evaluate the mean herbaceous biomass found in all three study sites and the mean woody plant biomass of each pit as we move down the catena. I will evaluate which soil particles are more suitable for good plant growth by looking at what happens to these factors as %clay and %sand increases or decreases in horizon A

Abstract
The report is based on trying to understand how particle size distribution in soil can affect the plants growing in that area. We know that the soil is where most life takes place, soil has six characteristics being texture, structure, bulk density, porosity, % soil organic matter and pH. All these characteristics are determined by the particle size distribution in the soil: sand 0.05-2.00mm, silt 0.002-0.05mm and clay greater than 0.002mm. For this experiment, students were taken to Tswaing crater to study the 220,000 ± 52,000 years old area and collect samples from three different soil pits located at the top, bottom and the middle of the catena. Soil samples and plant samples were collected and taken back to the lab for further lab analysis. With the samples collected a set of data was obtained and calculations using the data collected were done to support the results given in this report. The experiment is about comparing how mean tree height, the maximum tree height, mean woody plant biomass and mean herbaceous biomass changes as we adjust from sandy soils to clay soils at the bottom of the catena. The results show that at the bottom of the catena we have more clay soil while at the top of the catena we have sandy soils. The results also show that at the top of the catena we will see woodier biomass than at the bottom of the catena, while at the bottom of the catena we will see more herbaceous biomass. With further analysis of the data an unexpected result of clay soils having taller trees than sandy soils is shown. The reason for this experiment is to educate farmers and gardeners on which type of soil to use when growing your vegetation; it is important to consider the type of crops you use on your soil.

Introduction
Particle size distribution analysis is the measurement of the size distribution of individual sediments or soil particles (clay, silt, and sand), which is used to understand pedogenesis (soil formation), to classify soil or to define texture (Staff, 1993). Soil texture is based on the proportion of silt 0.05-0.002 mm, sand 2.0-0.05 mm, and clay less than 0.002 mm particles. From Table 1, geologists and sedimentologists have been at the forefront of studying submerged sediments and marine environments. Particle size distribution characteristics determine measures such as air-water relations, pore size distribution and water movement in the profile. Soil particle size distribution is

one of the most important soil physical attributes; it is of great importance to soil water movement (Fuqiang, 2011). Soil porosity is the space between soil particles, which has a varying amount of water and air. Porosity depends on both soil texture and structure. Water moves down the soil by gravity through open pore spaces in the soil; the size of the space and of soil particles determines how much water can flow in (Connor, 2014). Wide pore spacing at the soil surface has a high rate of water infiltration, so coarse soils will have a higher infiltration rate than fine soils. Soil is a valuable resource that supports plant life. Management decisions made concerning the types of crops to plant in soils, number of plants, watering scheduling, and the amount of nitrogen fertilizer to apply all depend on the amount of water in the soil that is available for the crop throughout the growing season. By understanding the physical characteristics of the soil, you will be better at defining the strengths and weaknesses of different soil types (Mancini, 2004). This information is very useful for farmers, researchers and people trying to grow their own garden at home. Particle size distribution also affects the type of nutrients that can be found in the soil, such as nitrogen, phosphorus, and potassium. For plants to grow in height, minerals like zinc need to be present in the soil. The particle size distribution will also impact the type of plants that will dominate the area because some plants need less water than other plants. The hypothesis for this experiment is that horizon A from each pit with a particle size distribution of %sand higher than %clay will have better plant growth, taller trees, mean herbaceous biomass and mean woody plant biomass being greater than in clay soils.

Material and methods

Description of site

Tswaing crater is about 25.4158° S, 28.1006° E and 50 km north of Pretoria. It is a sub-tropical area with dry and humid conditions, situated in the Savannah biome. The site is believed to be 220,000 ± 52,000 years old, well known for the meteorite of about 30 to 50 m that crashed there. Upon impact, the stony meteorite was vaporised during the impact event. Tswaing means place of salt. To really determine how plant growth changes in an area, three study sites were chosen: one from the top of the catena, one from the middle of the catena and the last at the bottom of the catena. Once at the top, middle and bottom of the catena, pick an area that has a clear field to dig a pit and enough plant species around to make observations. This is how the field work was done:

Figure 1: Map of Tswaing crater with red dots and red lines showing where the study sites were on the catena (top of the catena, middle of the catena, and bottom of the catena).

Field work

Soil pits of 1.5 m depth, 1.2 m deep and 0.8 m wide were dug or until hard rock was exposed. Before describing or photographing the profile, one face of the soil pit was freshly cleaned so that it was smooth and vertical, and all the horizons were in clear view. The location of the pit was taken with a Garmin GPS and written down on a paper bag that would hold soil samples. By looking at the soil

pit, the parent material is easily identified. Horizon A, B1 and B2 from each soil pit were identified and the depth of each was taken. Using the shovel, aggregate samples of about 1 kg of each horizon were collected, and the samples from each horizon were placed into three separate paper bags labelled with the name of the horizon and location of the pit. The samples were transported from the field in a cooler box and analysed immediately or air-dried. From each horizon a clod of the soil was collected and taken downhill to the van where a string was wrapped around it with a label tag attached to the string. The clod was dipped in hot wax three times and left out to dry. To measure vegetation, take the location of the study site with a Garmin GPS and lay out plots/ quadrats of known size. The size of the plot depends on the size of the vegetation in our study site; plots can range from 10cm x 10cm to 100m x 100m in a square shape. For the herbaceous layer, we will be placing contiguous quadrats, using a square quadrat of 0.5m x 0.5m in a long line, next to each other. This makes the quadrats not independent of one another. Take readings of bare but porous soil, faunal holes, cover of plant litter or dung, and basal area of rooted plants. Samples for woody species are taken in this order: canopy cover, which is the dominance in the specific study area. Lay out a 20m rod from the centre of the soil pit. For each 5m from the last study site, record all woody plant species you come across that are perpendicular to the transect. All the species recorded are given codes and the shape of their leaves is written down (they can be fine-leafed or broad-leafed). Looking around the transect, spot any insects or mammalian herbivory disturbance occurring and the results are taken down. The basal circumference and the height in cm of the tree or shrub species were measured using a measuring tape.

Lab analysis

To determine the particle size distribution, 50g of soil was placed into a 1-litre polythene bottle, and 100ml of dispersing agent (4% sodium hexametaphosphate) was added to the 50g of soil in the bottle. Shake the bottle mixture for 10 mins by hand, with my hand placed at the top of the bottle and at the bottom of the bottle. Soil and solution were transferred to a cup, washing out the bottle and making the volume in the cup up to about 500ml with water. Vigorous whisking of the solution for 2 minutes follows. Transfer the dispersed soil suspension into a 1000ml measuring cylinder and wash out the stirrer cup for any soil particles left behind. The volume of the cylinder was adjusted to 1 litre with water, the solution was mixed, and the first temperature reading was taken. Use a special plunger to mix by moving it up and down in the cylinder about 10 times. A timer was started and after the selected interval, the hydrometer was lowered carefully into the centre of the suspension and released gently. Take the reading on the scale and round it off to the nearest 0.5 unit. When bits of floating debris obscure the shaft of the hydrometer, a single drop of isopropyl alcohol was added to the surface of the water a second before taking the reading. The mixing process of the suspension is repeated, and we took the next reading after the indicated interval. Readings are taken at 20 seconds, 4 minutes and 2 hours.

Data analysis

The proportional amounts (in %clay and %sand) of the fractions are calculated by using the following formulas:

% clay = (C + TCF) * (WCF / m) * 100

% silt = (B + TCF) * (WCF / m) * 100 - (%clay)

% silt = (A + TCF) * (WCF / m) * 100 - (%clay + %silt(2 - 20))

% sand = 100 - (%clay + %silt(2 - 20) + %silt(20 - 50))

With TCF being the temperature correction factor obtained from a table, WCF is the moisture correction factor = (100 + %moisture) / 100. The value is 1 if the soil is oven-dry, and m is the dry mass of soil measured in grams. From the data of tree height, an average was taken of all the tree heights by adding all the heights and dividing by how many trees there were. To quantify the above-ground biomass for each tree/shrub using Nicklas's equation, convert back to mass by: m = exp(9 - 3.47 + 2.83 * Ln(d)).

Results

VEG DATA	bottom	middle	top
% Sand	36,0	66,0	69,4
% Clay	31,6	17,0	12,9
Mean herbaceous biomass (tons/ha)	2,54	1,54	1,36
Mean woody biomass (kg)	0,05	0,06	0,07
Mean tree height (m)	1,51	1,92	1,80
Mean maximum tree height (m)	1,93	2,95	3,47

Table 1: Table of results of physical growth of plants from each soil pit and particle size distribution from horizon A of each soil pit down the catena.

Figure 2: Line graph of how the particle size distribution of horizon A changes as we move down the catena. The orange line represents % clay. From the graph, we see that as we move down the catena % clay decreases from 31,6 to 12,9. For % sand the opposite can be said as it increases as you move down the catena; it increases from 36,0 to 69,4.

Figure 3: Line graph showing how a change in % sand in horizon A affects the physical growth of plants from each soil pit down the catena. Mean herbaceous biomass decreases down the catena, mean woody biomass increases down the catena, mean tree height increases down the catena and lastly mean maximum tree height increases down the catena. All these factors are taken and compared as the %sand increases from soil pit one to soil pit 3.

Figure 4: Line graph showing how a change in % clay in horizon A affects the physical growth of plants from each soil pit down the catena. Mean herbaceous biomass decreases down the catena, mean woody biomass increases down the catena, mean tree height increases down the catena and lastly mean maximum tree height increases down the catena. All these factors are taken and compared as the % clay decreases from soil pit one to soil pit 3.

Discussion

%Clay decreases as you move down the catena. Clays are known to have the following characteristics: Clay soils generally feel very sticky when wet. They can hold more total water than most other soil types because they have a higher field capacity. Because of this, only about half of this water is available to plants. They swell when saturated with water and shrink when dried; this is due to them having small pore spaces, so a certain amount of restructuring can take place in these soils depending on weather conditions (Finch, 2014). They warm up late in the spring because water heats up more slowly than mineral matter. They are normally fairly rich in potash which helps plants uptake nitrogen from the soil, but are not rich in phosphates which are used by plants during photosynthesis and respiration. % sand increases as you move down the catena. Sandy soils have a light texture and are known to have a loose structure, which causes them to drain water quickly and not be able to hold water well (Fry, 2013). They have bigger pore spaces; this leads them to have a lower field capacity. This will lead to low fertility, but with the addition of some fertilising products, sandy soil can grow a satisfactory garden. This type of soil will dry quickly. Herbaceous biomass comes from plants that do not have a woody stem and die back at the end of their growing season (Elbersen, 2017). They have annual, biennial and perennial life cycles. The herbaceous biomass decreases as you move down the catena. This means that more herbaceous biomass is found at the top of the catena; this also shows that herbaceous plants are prone to grow better in sandy soils than in clay soil. Because of the large particles, herbaceous plant roots can easily penetrate the soil; the soil is porous and more malleable. Most herbaceous plants tend to tolerate the dryness of sandy

soils. Because sandy soils warm up early in spring, this makes them well suited for herbaceous plants as they thrive better in spring. Herbaceous plants that grow better in sandy soils include lettuce, collard greens, moss phlox and Schmidtia pappophoroides. The opposite can be said for woody plants; woody plants are usually trees, lianas or shrubs. Woody plant biomass decreases as you move down the catena, which means that woody plants grow better in clay soils than in sandy soils. Trees have more shallow roots; this increases their wilting point, meaning that they can extract more water out of the soil before the water is unable to be extracted. Often woody plants have taproots, and their taproots help them break up tough clay soils. Because of clay soils' high field capacity, they can hold more water and nutrients, leaving trees with more nutrients to grow strong stems. Woody plants are bigger and therefore require more nutrients and water than herbaceous plants. Plants such as fragrant bayberries, dogwoods, viburnums and Aristida congesta will grow better in clay soils as they are more adapted to the conditions of clay soils. An experiment was done in Australia to investigate the role of clay on the development of woody biomass in drylands found in Australia. The results showed substantially lower woody biomass growth on clay soils than non-clay soils in drylands found in Australia (Butler, 2015). These results argue with the results I have obtained at Tswaing crater in Pretoria; the results I obtained show that woody biomass is higher in clay soils than it is in non-clay soils. The height of a tree is known as the number of edges on the longest downward path between the leaf and roots of the tree. From the results we see that as you move down the catena tree height increases, meaning that tall trees are more likely to grow in sandy soils than in clay soils. Because sandy soils have larger pores, trees that grow in the soil will have more water to use as it has a lower field capacity. But for clay soils, even though the soil can hold on to the water, plants will not be able to get the water because of its high field capacity. As we know, trees need water to grow tall. Primary macronutrients that trees need to thrive are nitrogen (which helps trees to convert water and sunlight into food), phosphorus (which is responsible for root growth) and lastly potassium (which protects trees from cold seasons). An experiment was done by Shermeen Tahir and his colleagues to compare nitrogen and phosphorus retention in sandy and clay soils. The results showed that eighty-three percent more nitrogen was retained in clay soils and phosphorus retention in clay soils was doubled. This experiment showed that clay soil can substantially reduce nitrogen and phosphorus leaching and ultimately increase the fertiliser retention of soil compared to sandy soil. The overall conclusion of this experiment is that herbaceous plants grow better in sandy soils, woody plants grow better in clay soils, and trees will grow taller in sandy soils than in clay soils. Two of my hypotheses were correct.

All the figures referred in this document are in the file called report figures.

Reference

Butler, D., June 2015. How does clay constrain woody biomass in drylands? ResearchGate, pp. 20.
Connor, S., August 2014. Influence of particle size distribution. ScienceDirect, pages 99-105.

Elbersen, W., 2017. Lignocellulosic Biomass Quality. ScienceDirect, pp. 103-107.
Finch, H.J.S., 2014. Soils and soil management. ScienceDirect, pp. 7.
Fuqiang, T., June 2011. Soil particle size distribution and its relationship with soil water and salt under mulched drip irrigation in Xinjiang of China. Science Technology, pp. 12-13.
Fry, M., 21 July, 2013. The Properties of Sandy Soils. ResearchGate, pp. 57.
Mancini, C., 2004. Soil and Water Relationships. Noble Research Institute, pp. 47-52.
Staff, B.L., 1993. Soil Survey Manual. Agricultural Handbook, USDA-NRCS, No. 18, pp. 234-245.